understand what a model is. We use this term informally to refer to a neural network, and it's often used to refer to any kind of statistical model, others being a linear model or a Bayesian graphical model. In another context, we might say a model is a mental or mathematical representation of how something works in "the real world." If we understand exactly how something works (i.e., what it's composed of and how those components interact) then we can not only explain data we've already seen, but we can predict data we haven't yet seen.

For example, weather forecasters build very sophisticated models of the climate that take into account many relevant variables, and they're constantly measuring real-world data. They can use their models to predict the weather to some degree of accuracy. There's an almost cliché statistics mantra that "all models are wrong, but some are useful," meaning that it is impossible to build a model that 100% corresponds to reality; there will always be data or relationships that we're missing. Nonetheless, many models capture enough truth about a system we're interested in that they're useful for explanation and prediction.

If we could build an algorithm that could figure out how Gridworld works, it would have inferred a model of Gridworld, and it would be able to play it perfectly. In Q-learning, all we gave the Q-network was a numpy tensor. It had no *a priori* model of Gridworld, but it still learned to play by trial and error. We did not task the Q-network with figuring out how Gridworld works; its only job was to predict expected rewards. Hence, Q-learning is a model-free algorithm.

As the human architects of algorithms, we may be able to engineer in some of our own domain knowledge about a problem as a model to optimize our problem. We could then supply this model to a learning algorithm and let it figure out the details. This would be a model-based algorithm. For example, most chess-playing algorithms are model-based; they know the rules of how chess works and what the result of taking certain moves will be. The only part that isn't known (and that we'd want the algorithm to figure out) is what sequence of moves will win the game. With a model in hand, the algorithm can make long-term plans in order to achieve its aim.